Common Characters

AlgoExpert · Easy · Strings

The problem

- Return the unique characters that appear in every string.

Intuition

- A character is common if it appears in every string, so count in how many strings each unique character appears.

Approach

- 1. For each string, take its set of characters
- 2. Count the number of strings containing each one
- 3. Keep characters whose count equals the number of strings

Java solution

```java
public class CommonCharacters {
    public static String[] commonCharacters(String[] strings) {
        Set<Character> common = null;
        for (String s : strings) {
            Set<Character> cur = new HashSet<>();
            for (char c : s.toCharArray()) cur.add(c);
            if (common == null) common = cur; else common.retainAll(cur);
        }
        List<String> out = new ArrayList<>();
        for (char c : new TreeSet<>(common)) out.add(String.valueOf(c));
        return out.toArray(new String[0]);
    }
}
```

Complexity

- Time: O(n*m)
- Space: O(c) distinct chars

Recap

- A character is common if it appears in every string, so count in how many strings each unique character appears.
- Complexity: O(n*m)